Project Idea Proposal: SunMap Syria

Group 12

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Machine Learning-Based Solar Potential Mapping and Next-Day Yield Forecasting across Syrian Governorates

1. Project Idea

- **Problem Statement:** Syria faces severe energy deficits and frequent power grid instability. While solar photovoltaics (PV) offer a viable clean energy alternative, the lack of localized empirical solar output data and real-time monitoring infrastructure hampers effective site selection, grid integration, and energy planning.
- **Specific Goal:** This project aims to build **SunMap Syria**, a machine learning pipeline that models daily photovoltaic potential (*kWh/kWp*) across all 14 Syrian governorates. The system evaluates spatial-temporal solar potential and provides a 24-hour ahead forecasting model using historical meteorological and satellite data.
- **Methodological Note:** Due to the absence of widespread physical smart-metering infrastructure in Syria, baseline PV yield data is derived from the validated PVGIS simulation model combined with NASA POWER atmospheric data.

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2. Relevance to Sustainable Development Goals (SDGs)

This project directly aligns with three United Nations Sustainable Development Goals:

- **SDG 7: Affordable and Clean Energy:** By quantifying annual yield potential (ranging from ~1,617 *kWh/kWp* in coastal regions to >1,760 *kWh/kWp* in inland areas), the project assists decision-makers and localized micro-grid developers in optimizing renewable infrastructure investments.
- **SDG 13: Climate Action:** Facilitating a data-driven transition from reliance on fossil-fuel generators to solar PV significantly reduces regional greenhouse gas emissions and carbon footprint.
- **SDG 11: Sustainable Cities and Communities:** Accurate 24-hour next-day yield forecasting enhances energy resilience, allowing local communities and utility operators to manage micro-grid stability amidst grid instability.

3. Literature Examples

1. **Kruitwagen et al. (2021) –** ***A global inventory of photovoltaic solar energy generating facilities***
    - *Summary & Relevance:* This study leveraged satellite imagery and machine learning to map global solar infrastructure. It provides inspiration for utilizing remote-sensing and geospatial dataset integration when ground-based measurement stations are unavailable.

2. **Yu et al. (2018) –** ***Deep learning for regional solar power forecasting using multi-source meteorological data***
    - *Summary & Relevance:* Yu et al. demonstrated that combining temporal atmospheric variables (irradiance, cloud index, relative humidity, and temperature) with gradient boosting and deep learning models markedly improves short-term PV output predictions compared to traditional persistence baselines.